Skill-2

1. **Create Table**

```
CREATE TABLE student_marks1 (
 roll_no INT PRIMARY KEY,
 name VARCHAR(50),
 subject VARCHAR(50),
 marks DECIMAL(5,2)
);
```

```
mysql> USE sec13;
Database changed
mysql> CREATE TABLE student_marks1 (
    ->     roll_no INT PRIMARY KEY,
    ->     name VARCHAR(50),
    ->     subject VARCHAR(50),
    ->     marks DECIMAL(5,2)
    -> );
Query OK, 0 rows affected (0.51 sec)
```

2. **Insert Data**

```
INSERT INTO student_marks1 (roll_no, name, subject,
marks) VALUES
(1, 'Ravi', 'Math', 85.50),
(2, 'Sita', 'Math', 92.75),
(3, 'Anil', 'Math', 78.40),
(4, 'Priya', 'Math', 88.90),
(5, 'Vijay', 'Math', 80.25),
(6, 'Ananya', 'aws', 98.50),
(7, 'Kumar', 'dbms', 95.75),
(8, 'Akhil', 'english', 97.40),
(9, 'Akhi', 'aws1', 99.90),
(10, 'Rudra', 'azure', 82.25);
```

```
mysql> USE sec13;
Database changed
mysql> INSERT INTO student_marks1 (roll_no, name, subject, marks) VALUES
    -> (1, 'Ravi', 'Math', 85.50),
    -> (2, 'Sita', 'Math', 92.75),
    -> (3, 'Anil', 'Math', 78.40),
    -> (4, 'Priya', 'Math', 88.90),
    -> (5, 'Vijay', 'Math', 80.25),
    -> (6, 'Ananya', 'aws', 98.50),
    -> (7, 'Kumar', 'dbms', 95.75),
    -> (8, 'Akhil', 'english', 97.40),
    -> (9, 'Akhi', 'aws1', 99.90),
    -> (10, 'Rudra', 'azure', 82.25);
Query OK, 10 rows affected (0.13 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

3. **Simple Select**

```
SELECT * FROM student_marks1;
```

```
mysql> SELECT * FROM student_marks1;
+---------+--------+---------+-------+
| roll_no | name   | subject | marks |
+---------+--------+---------+-------+
|       1 | Ravi   | Math    | 85.50 |
|       2 | Sita   | Math    | 92.75 |
|       3 | Anil   | Math    | 78.40 |
|       4 | Priya  | Math    | 88.90 |
|       5 | Vijay  | Math    | 80.25 |
|       6 | Ananya | aws     | 98.50 |
|       7 | Kumar  | dbms    | 95.75 |
|       8 | Akhil  | english | 97.40 |
|       9 | Akhi   | aws1    | 99.90 |
|      10 | Rudra  | azure   | 82.25 |
+---------+--------+---------+-------+
10 rows in set (0.02 sec)
```

4. **Aggregate Functions**

```
SELECT COUNT(*) AS total_students FROM student_marks1;
SELECT SUM(marks) FROM student_marks1;
SELECT AVG(marks) FROM student_marks1;
SELECT MAX(marks) FROM student_marks1;
SELECT MIN(marks) FROM student_marks1;
```

```
mysql> SELECT COUNT(*) AS total_students FROM student_marks1;
+----------------+
| total_students |
+----------------+
|             10 |
+----------------+
1 row in set (0.01 sec)

mysql> SELECT SUM(marks) FROM student_marks1;
+------------+
| SUM(marks) |
+------------+
|     899.60 |
+------------+
1 row in set (0.01 sec)

mysql> SELECT AVG(marks) FROM student_marks1;
+------------+
| AVG(marks) |
+------------+
|  89.960000 |
+------------+
1 row in set (0.00 sec)

mysql> SELECT MAX(marks) FROM student_marks1;
+------------+
| MAX(marks) |
+------------+
|      99.90 |
+------------+
1 row in set (0.01 sec)

mysql> SELECT MIN(marks) FROM student_marks1;
+------------+
| MIN(marks) |
+------------+
|      78.40 |
+------------+
1 row in set (0.00 sec)
```

5. **Filtering with WHERE**

```
SELECT * FROM student_marks1 WHERE marks > 85;
SELECT * FROM student_marks1 WHERE marks >= 90;
SELECT * FROM student_marks1 WHERE marks < 80;
SELECT * FROM student_marks1 WHERE marks BETWEEN 80
AND 90;
SELECT * FROM student_marks1 WHERE name LIKE 'P%';
```

```
SELECT * FROM student_marks1 WHERE name IN ('Ravi',
'Sita', 'Vijay');
SELECT * FROM student_marks1 WHERE marks > 85 AND
(subject = 'Math' OR name LIKE 'P%');
```

```
mysql> SELECT * FROM student_marks1 WHERE marks > 85;
+---------+--------+---------+-------+
| roll_no | name   | subject | marks |
+---------+--------+---------+-------+
|       1 | Ravi   | Math    | 85.50 |
|       2 | Sita   | Math    | 92.75 |
|       4 | Priya  | Math    | 88.90 |
|       6 | Ananya | aws     | 98.50 |
|       7 | Kumar  | dbms    | 95.75 |
|       8 | Akhil  | english | 97.40 |
|       9 | Akhi   | aws1    | 99.90 |
+---------+--------+---------+-------+
7 rows in set (0.02 sec)

mysql> SELECT * FROM student_marks1 WHERE marks >= 90;
+---------+--------+---------+-------+
| roll_no | name   | subject | marks |
+---------+--------+---------+-------+
|       2 | Sita   | Math    | 92.75 |
|       6 | Ananya | aws     | 98.50 |
|       7 | Kumar  | dbms    | 95.75 |
|       8 | Akhil  | english | 97.40 |
|       9 | Akhi   | aws1    | 99.90 |
+---------+--------+---------+-------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM student_marks1 WHERE marks < 80;
+---------+------+---------+-------+
| roll_no | name | subject | marks |
+---------+------+---------+-------+
|       3 | Anil | Math    | 78.40 |
+---------+------+---------+-------+
1 row in set (0.00 sec)
```

```
mysql> SELECT * FROM student_marks1 WHERE marks BETWEEN 80 AND 90;
+---------+-------+---------+-------+
| roll_no | name  | subject | marks |
+---------+-------+---------+-------+
|       1 | Ravi  | Math    | 85.50 |
|       4 | Priya | Math    | 88.90 |
|       5 | Vijay | Math    | 80.25 |
|      10 | Rudra | azure   | 82.25 |
+---------+-------+---------+-------+
4 rows in set (0.01 sec)

mysql> SELECT * FROM student_marks1 WHERE name LIKE 'P%';
+---------+-------+---------+-------+
| roll_no | name  | subject | marks |
+---------+-------+---------+-------+
|       4 | Priya | Math    | 88.90 |
+---------+-------+---------+-------+
1 row in set (0.01 sec)

mysql> SELECT * FROM student_marks1 WHERE name IN ('Ravi', 'Sita', 'Vijay');
+---------+-------+---------+-------+
| roll_no | name  | subject | marks |
+---------+-------+---------+-------+
|       1 | Ravi  | Math    | 85.50 |
|       2 | Sita  | Math    | 92.75 |
|       5 | Vijay | Math    | 80.25 |
+---------+-------+---------+-------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM student_marks1 WHERE marks > 85 AND (subject = 'Math' OR name LIKE 'P%');
+---------+-------+---------+-------+
| roll_no | name  | subject | marks |
+---------+-------+---------+-------+
|       1 | Ravi  | Math    | 85.50 |
|       2 | Sita  | Math    | 92.75 |
|       4 | Priya | Math    | 88.90 |
+---------+-------+---------+-------+
3 rows in set (0.00 sec)
```

6.Update Queries

```
UPDATE student_marks1 SET marks = 90.00 WHERE roll_no = 3;
UPDATE student_marks1 SET subject = 'Remedial Math' WHERE marks < 80;
```

```
mysql> UPDATE student_marks1 SET marks = 90.00 WHERE roll_no = 3;
Query OK, 1 row affected (0.03 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE student_marks1 SET subject = 'Remedial Math' WHERE marks < 80;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0
```

7.Delete Queries

```
DELETE FROM student_marks1 WHERE roll_no = 5;
DELETE FROM student_marks1 WHERE marks < 75;
```

```
mysql> DELETE FROM student_marks1 WHERE roll_no = 5;
Query OK, 1 row affected (0.01 sec)

mysql> DELETE FROM student_marks1 WHERE marks < 75;
Query OK, 0 rows affected (0.00 sec)
```

8. Sorting

```
SELECT * FROM student_marks1 ORDER BY marks ASC;
SELECT * FROM student_marks1 ORDER BY marks DESC;
SELECT * FROM student_marks1 ORDER BY name ASC;
SELECT * FROM student_marks1 ORDER BY name DESC;
```

```
mysql> SELECT * FROM student_marks1 ORDER BY marks ASC;
+---------+--------+---------+-------+
| roll_no | name   | subject | marks |
+---------+--------+---------+-------+
|      10 | Rudra  | azure   | 82.25 |
|       1 | Ravi   | Math    | 85.50 |
|       4 | Priya  | Math    | 88.90 |
|       3 | Anil   | Math    | 90.00 |
|       2 | Sita   | Math    | 92.75 |
|       7 | Kumar  | dbms    | 95.75 |
|       8 | Akhil  | english | 97.40 |
|       6 | Ananya | aws     | 98.50 |
|       9 | Akhi   | aws1    | 99.90 |
+---------+--------+---------+-------+
9 rows in set (0.01 sec)

mysql> SELECT * FROM student_marks1 ORDER BY marks DESC;
+---------+--------+---------+-------+
| roll_no | name   | subject | marks |
+---------+--------+---------+-------+
|       9 | Akhi   | aws1    | 99.90 |
|       6 | Ananya | aws     | 98.50 |
|       8 | Akhil  | english | 97.40 |
|       7 | Kumar  | dbms    | 95.75 |
|       2 | Sita   | Math    | 92.75 |
|       3 | Anil   | Math    | 90.00 |
|       4 | Priya  | Math    | 88.90 |
|       1 | Ravi   | Math    | 85.50 |
|      10 | Rudra  | azure   | 82.25 |
+---------+--------+---------+-------+
9 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM student_marks1 ORDER BY name ASC;
+---------+--------+---------+-------+
| roll_no | name   | subject | marks |
+---------+--------+---------+-------+
|       9 | Akhi   | aws1    | 99.90 |
|       8 | Akhil  | english | 97.40 |
|       6 | Ananya | aws     | 98.50 |
|       3 | Anil   | Math    | 90.00 |
|       7 | Kumar  | dbms    | 95.75 |
|       4 | Priya  | Math    | 88.90 |
|       1 | Ravi   | Math    | 85.50 |
|      10 | Rudra  | azure   | 82.25 |
|       2 | Sita   | Math    | 92.75 |
+---------+--------+---------+-------+
9 rows in set (0.01 sec)

mysql> SELECT * FROM student_marks1 ORDER BY name DESC;
+---------+--------+---------+-------+
| roll_no | name   | subject | marks |
+---------+--------+---------+-------+
|       2 | Sita   | Math    | 92.75 |
|      10 | Rudra  | azure   | 82.25 |
|       1 | Ravi   | Math    | 85.50 |
|       4 | Priya  | Math    | 88.90 |
|       7 | Kumar  | dbms    | 95.75 |
|       3 | Anil   | Math    | 90.00 |
|       6 | Ananya | aws     | 98.50 |
|       8 | Akhil  | english | 97.40 |
|       9 | Akhi   | aws1    | 99.90 |
+---------+--------+---------+-------+
9 rows in set (0.00 sec)
```

9. Group By

```
SELECT subject, SUM(marks) AS total_marks FROM
student_marks1 GROUP BY subject;
SELECT subject, AVG(marks) AS avg_marks FROM
student_marks1 GROUP BY subject;
SELECT subject, COUNT(*) AS num_students FROM
student_marks1 GROUP BY subject;
```

```
mysql> SELECT subject, SUM(marks) AS total_marks FROM student_marks1 GROUP BY subject;
+---------+-------------+
| subject | total_marks |
+---------+-------------+
| Math    |      357.15 |
| aws     |       98.50 |
| dbms    |       95.75 |
| english |       97.40 |
| aws1    |       99.90 |
| azure   |       82.25 |
+---------+-------------+
6 rows in set (0.03 sec)

mysql> SELECT subject, AVG(marks) AS avg_marks FROM student_marks1 GROUP BY subject;
+---------+-----------+
| subject | avg_marks |
+---------+-----------+
| Math    | 89.287500 |
| aws     | 98.500000 |
| dbms    | 95.750000 |
| english | 97.400000 |
| aws1    | 99.900000 |
| azure   | 82.250000 |
+---------+-----------+
6 rows in set (0.01 sec)

mysql> SELECT subject, COUNT(*) AS num_students FROM student_marks1 GROUP BY subject;
+---------+--------------+
| subject | num_students |
+---------+--------------+
| Math    |            4 |
| aws     |            1 |
| dbms    |            1 |
| english |            1 |
| aws1    |            1 |
| azure   |            1 |
+---------+--------------+
6 rows in set (0.00 sec)
```

10. Having

```
SELECT subject, AVG(marks) AS avg_marks
FROM student_marks1
GROUP BY subject
HAVING AVG(marks) > 90;

SELECT subject, COUNT(*) AS num_students
FROM student_marks1
GROUP BY subject
HAVING COUNT(*) > 1;
```

```
mysql> SELECT subject, AVG(marks) AS avg_marks
    -> FROM student_marks1
    -> GROUP BY subject
    -> HAVING AVG(marks) > 90;
+---------+-----------+
| subject | avg_marks |
+---------+-----------+
| aws     | 98.500000 |
| dbms    | 95.750000 |
| english | 97.400000 |
| aws1    | 99.900000 |
+---------+-----------+
4 rows in set (0.01 sec)
```

```
mysql> SELECT subject, COUNT(*) AS num_students
    -> FROM student_marks1
    -> GROUP BY subject
    -> HAVING COUNT(*) > 1;
+---------+--------------+
| subject | num_students |
+---------+--------------+
| Math    |            4 |
+---------+--------------+
1 row in set (0.00 sec)
```